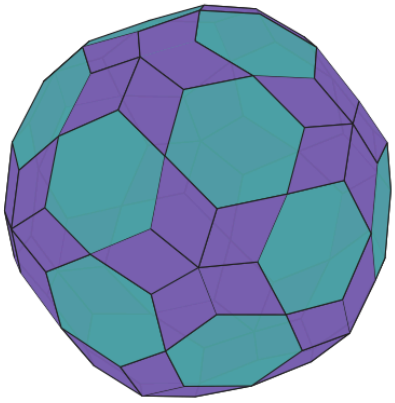
Picture book: new figures, 5 November 2026

DICTO · dictospheres.com · for looking, not for publication. Each picture's home is in the Concordance (destinations, 5 November 2026).

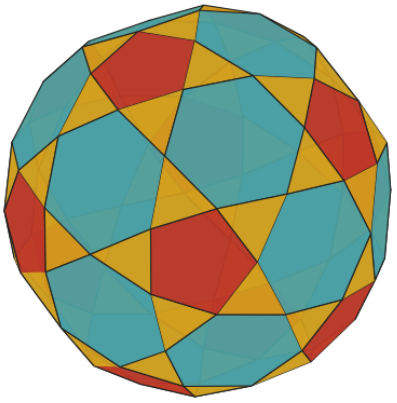
1. Pentagons in the OG

DICTO's proposal: a pentagon in the middle of each five-kite star. One flat cut at each of the 12 tips; what remains is the rectified truncated icosahedron. Destination: OG paper v5.

DICTO Ogdoecontahedron (OG)
20 hexagons + 60 kites



OG with its 12 five-fold tips cut
= rectified truncated icosahedron
20 hexagons, 12 pentagons, 60 triangles

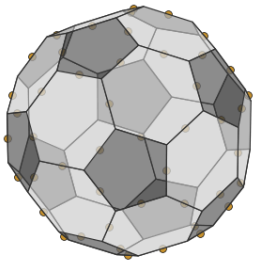


The 90 remaining corners all lie on the OG sphere (r = 2/3): the edge midpoints of a regular truncated icosahedron.
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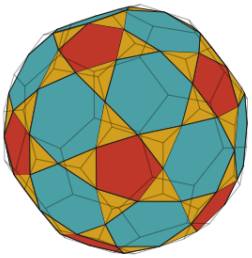
2. The football inside

The 90 corners of the cut OG are exactly the edge midpoints of the truncated icosahedron (C60's bond centres); right: the all-equal-edge version. Destination: OG paper v5.

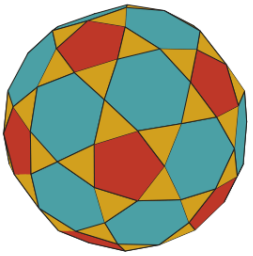
1. The football (truncated icosahedron, C_{60}) with the midpoints of its 90 edges



1. The OG with its tips cut: its 90 corners are exactly those midpoints (football edges shown)



2. All 180 edges equal: equilateral triangles, regular pentagons, hexagons 124.7° / 115.3°

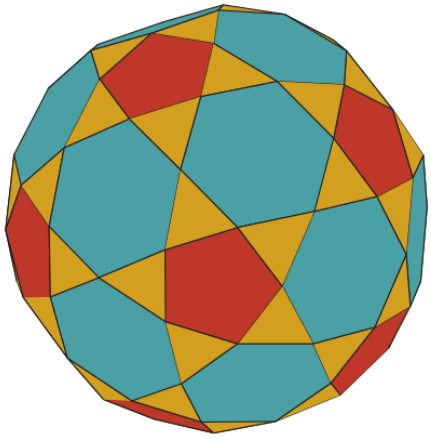


Pentagons red, hexagons teal, triangles gold. On the right the corners lie on two spheres (radius ratio 1.006); in the middle on one (r = 2/3).
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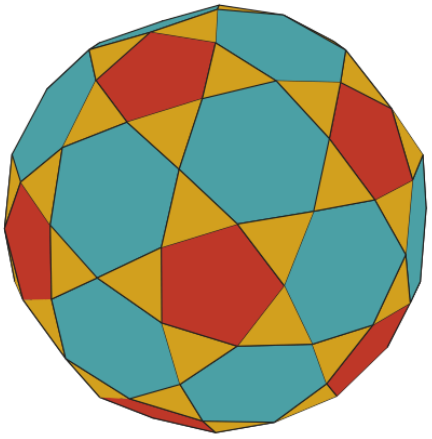
3. Regular hexagons or equilateral triangles

Same faces, two versions: on one sphere with regular hexagons, or all edges equal with equilateral triangles. Never both. Destination: OG paper v5.

OG with tips cut (one sphere)
regular hexagons, triangles 55.7° / 62.2° / 62.2°



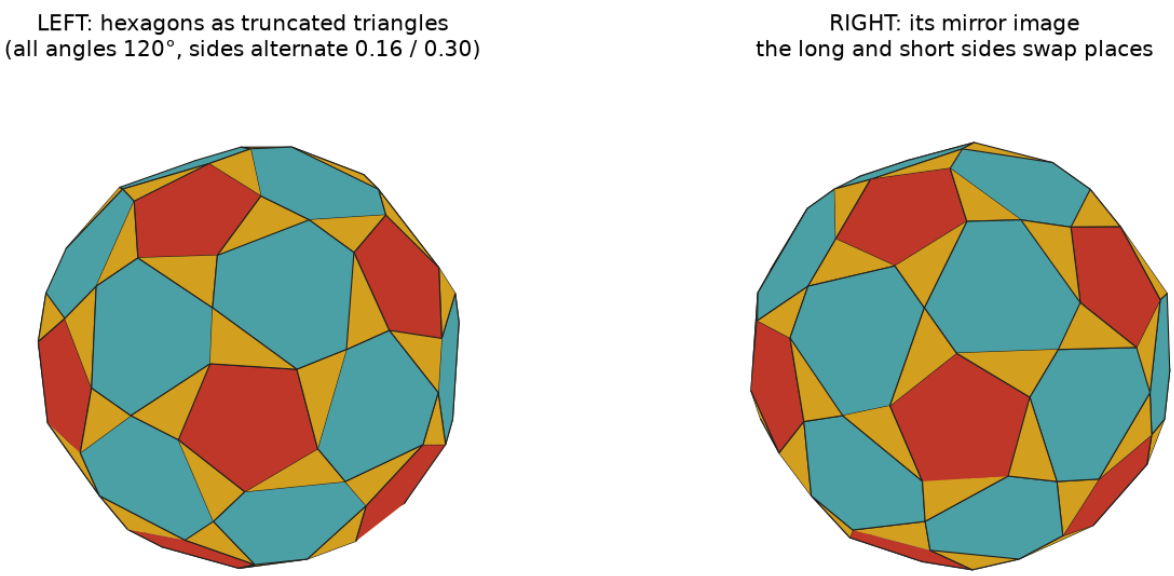
All 180 edges equal
equilateral triangles, regular pentagons, hexagons with angles 124.7° / 115.3°



Same faces, two choices: regular hexagons or equilateral triangles, never both.
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4. Hexagons as truncated triangles

DICTO's proposal: possible only by giving up mirror symmetry, so the solid comes as a left and a right. Destination: OG paper v5.



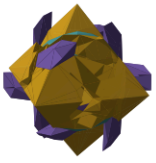
Hexagons as truncated triangles: possible only by giving up mirror symmetry, so the solid comes as a left and a right.
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5. The cut OG in the octet truss

Four shapes and a left/right pair fill the octet: cut OG, pentagonal connector, corner piece, dented tetrahedra. The filled octahedron below is hard to read at this size. Destination: Bull-horns octet paper, next version.



one octahedron of the octet, filled: cut OG (teal), connectors (purple), 6 corner pieces (gold)

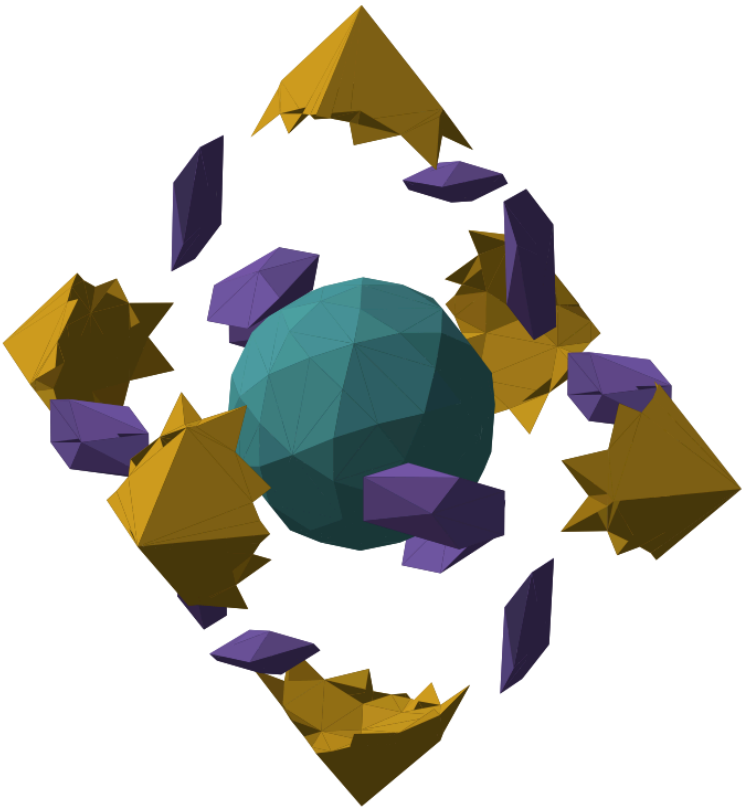


Cut-OG octet kit: 4 shapes and a left/right pair; volumes add up to an octahedron and two tetrahedra. DICTO · dictospheres.com · 5 Nov 2026

5b. One octahedron, exploded

The cut OG at the centre, the 12 pentagonal connectors on its edges (each shared with the neighbour across that edge) and the 6 corner pieces, all one shape.

One octahedron of the cut-OG octet kit, exploded
cut OG (teal) · 12 pentagonal connectors on its edges (purple) · 6 corner pieces (gold)

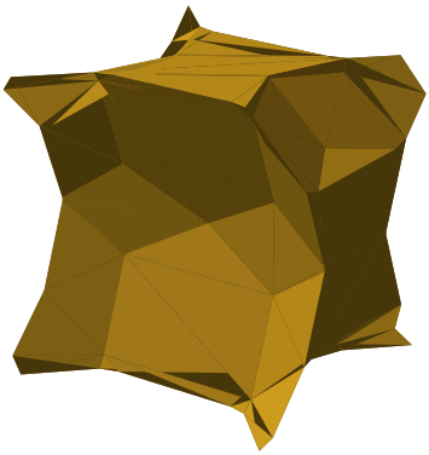


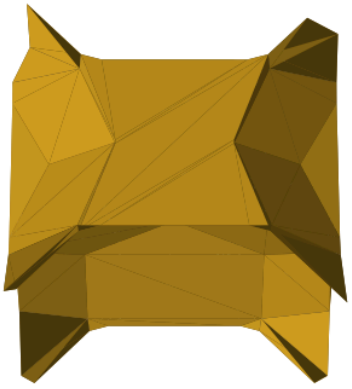
Each connector is shared with the neighbouring octahedron across that edge. DICTO · dictospheres.com · 5 Nov 2026

6. DICTO's Bull-horns adaptor, for the cut OG

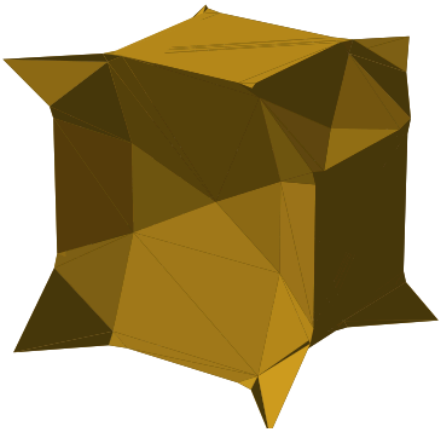
The cut OG and one new adaptor fill space with two shapes (76.16% / 23.84%). Destination: Golden Bodhi Pyrochlore paper, next version.

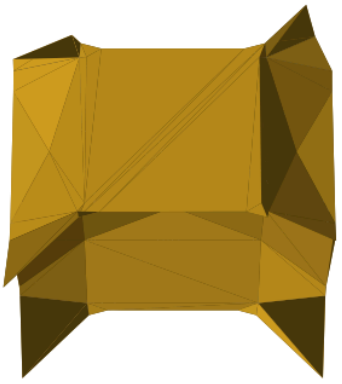
Bull-horns OG adaptor (original OG)
volume 0.111898, 22.57% of space





NEW: Bull-horns adaptor for the cut OG (pentagons)
volume 0.118230, 23.84% of space





Each piece fills one hole between 6 OGs of the body-centred cubic double network; OGs and adaptors fill space with two shapes.
Top: oblique view. Bottom: from above. DICTO · dictospheres.com · 5 Nov 2026